

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.04.

1 Find the Missing Angle

A triangle has two known angles: 48° and 65° .

Find the third angle.

$$180^\circ - 48^\circ - 65^\circ = 67^\circ \quad \underline{\hspace{2cm}}$$

2 Right Triangle Missing Angle

A right triangle has one angle that measures 37° .

Find the measure of the third angle.

$$180^\circ - 90^\circ - 37^\circ = 53^\circ \quad \underline{\hspace{2cm}}$$

3 Isosceles Triangle

An isosceles triangle has a vertex angle of 40° . The two base angles are equal.

Find the measure of each base angle.

$$(180^\circ - 40^\circ) \div 2 = 140^\circ \div 2 = 70^\circ \text{ each. } \underline{\hspace{2cm}}$$

4 Multi-Step

Two angles of a triangle are in the ratio 2:3. The third angle is 60° .

Find the measure of each of the two equal-ratio angles. (Hint: the two angles together = 120° .)

120° total; ratio 2:3 gives 48° and 72° . Check: $48^\circ + 72^\circ + 60^\circ = 180^\circ$.

5 Reasoning: Leo's Mistake

Leo found the angles of a triangle as 95° , 52° , and 43° . He says it is an obtuse triangle.

Check his arithmetic and his classification. Is he correct?

$95^\circ + 52^\circ + 43^\circ = 190^\circ \neq 180^\circ$. Leo made an arithmetic error. The correct third angle = $180^\circ - 95^\circ - 52^\circ = 33^\circ$, not 43° .